

WASP-43b

R-1513

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_190413 · created 2026-08-01T16:08:32+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458586.7148 +/- 0.0008 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.014
Transit depth (Rp/Rs)^2	3.52 +/- 0.53 [%]
Orbital Inclination (inc)	80.96 +/- 0.91
Airmass coefficient 1 (a1)	0.943 +/- 0.013
Airmass coefficient 2 (a2)	0.042 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	#1 - [69, 73]
Optimal Aperture	3.86
Optimal Annulus	9.7
Transit Duration (day)	0.049 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.188 ± 0.014 vs 0.1594 (deviation 2.01σ)
Duration fit vs published	0.049 d vs 0.0512 d (deviation 0.63σ)
Mid-time O–C	-4.2 min
Depth SNR	13.2
Residual scatter	1.36 %

Light curve

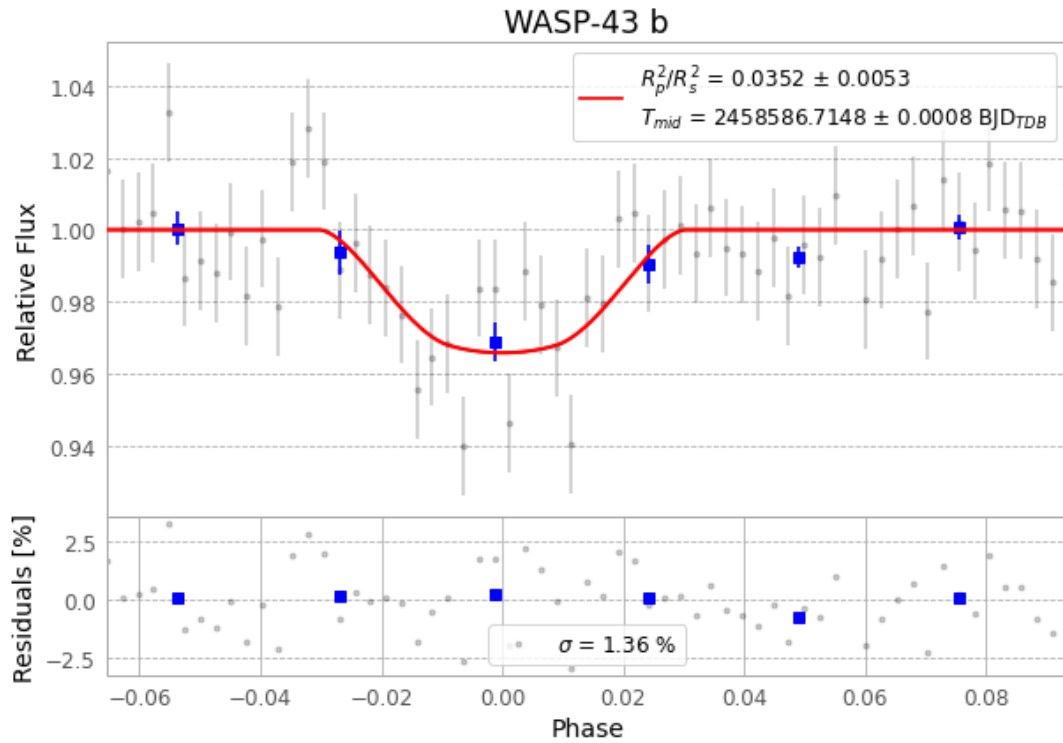


Figure 1 — FinalLightCurve_WASP-43 b_2019-04-12.png (SHA-256 77afdf63b3258a90...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2278523c4b5b0596...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), WASP-43190413034512.FITS → WASP-43190413065112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190413.log (66ed04a0c4b0...), FinalLightCurve_WASP-43 b_2019-04-12.png (77afdf63b325...), FinalLightCurve_WASP-43 b_2019-04-12.csv (18630911b22b...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 16:08Z.